A bipartite graph with a nonsplit action on its parts

Finite construction and elementary certificate

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Abstract

We give an explicit connected cubic bipartite graph whose automorphisms preserving its parts act transitively on edges, but whose automorphism group does not split over that subgroup. The graph and two permutations are specified by complete finite data. An elementary certificate proves that these permutations generate the full automorphism group, so the absence of part-exchanging involutions is not merely a property of a chosen subgroup.

1 The construction and its finite data

Let B be the graph on $\{1, \dots, 896\}$ with adjacency lists in `graph.json`. A row lists exactly the three neighbours of the vertex with that row number. The same file gives two permutations by their complete image lists. All indices in this data file start at one. These lists, rather than a random search or a chosen graph-library labelling, define the graph.

Two whole files form the required certificate: `check.py` and `graph.json`. Only Python's standard library is used. The successful report, `certificate.json`, binds both files by SHA256. The checks use explicit conditional failures and remain active under Python's `-O` option. The certificate does not call GAP, nauty, Bliss, or a graph isomorphism package.

Theorem 1. *The graph B is connected, simple, cubic and bipartite, with parts P, Q of size* 448 *and* 1344 *edges. Write $A = \mathrm{Aut}(B)$ and let A^+ be the subgroup preserving each part. Then*

$$|A| = 5376, \qquad |A^+| = 2688,$$

and A^+ is transitive on the edges. The elements of $A \setminus A^+$ have the following orders and multiplicities:

order	4	12	28
multiplicity	1024	896	768.

In particular, the sequence $1 \longrightarrow A^+ \longrightarrow A \longrightarrow C_2 \longrightarrow 1$ does not split.

The proof below separates the small general argument certifying fullness from the finite calculations. The latter are required premises and their whole data and implementation are retained. Optional independent computations with nauty and Bliss agree, but are not needed for the elementary proof.

2 Why the computed group is the full group

For a finite graph and an initial assignment of integer colours, replace the colour of each vertex by the pair consisting of its previous colour and the multiset of the previous colours of its neighbours. Encode equal pairs by equal new integers, using a fixed sorted order. This is neighbour-colour refinement. It never merges two previous colour classes.

Lemma 2. *Let X be a finite d-regular graph on n vertices. Suppose that giving distinct colours to one vertex v and its d neighbours, and one additional colour to all other vertices, eventually produces singleton classes under neighbour-colour refinement. Then* $|\mathrm{Aut}(X)| \le nd!$.

Proof. An automorphism fixing v and every neighbour of v preserves the initial colours. Inductively it preserves each refined colour: it preserves both the old colour and the multiset of old colours in the neighbourhood. When all colour classes are singletons, the automorphism fixes every vertex. Consequently an automorphism is determined by its action on the ordered star consisting of v and its neighbours. There are at most n choices for the image of v and $d!$ bijections from its neighbourhood to that of the image. This proves the bound. □

Proof of Theorem 1. The certificate first checks that every adjacency row contains three distinct valid vertices, excludes its own vertex, and is reciprocal. Breadth-first traversal visits all 896 vertices, assigns opposite binary colours across every edge and finds 448 vertices of each colour. Counting unordered adjacent pairs gives 1344 edges.

Individualize vertex 1 and its three neighbours. Starting with five colour classes, successive strict refinement steps give

$$5, 8, 11, 14, 17, 20, 26, 44, 135, 398, 692, 842, 890, 896.$$

The final partition is discrete. Lemma 2 therefore gives $|A| \le 896 \cdot 6 = 5376$.

The two supplied permutations are bijections of the vertex set and preserve every adjacency row. Their orders are 6 and 4. Starting at the identity, the certificate closes a set of permutations under multiplication by these two generators and obtains exactly 5376 distinct permutations. This finite closed monoid of permutations is their generated group: every inverse is a positive power in the finite permutation group. It is contained in A and attains the upper bound, so it equals A.

For every one of these 5376 permutations, the certificate tests its image of P. Exactly 2688 preserve P, and the other 2688 send it to Q. The orbit of one unordered edge under the former permutations consists of all 1344 edges. For each of the remaining permutations its order is computed as the least common multiple of the lengths of its disjoint cycles. The resulting distribution is the one displayed in the theorem; its entries sum to 2688.

The part action is onto C_2, with kernel A^+. A section of this action would send the nonidentity element of C_2 to an involution outside A^+. The exhaustive distribution contains no such element. Conversely such an involution would generate a complement, so the stated obstruction is exact. □

3 Scope and reproducibility

The graph supplies an input for a separate construction of a thin flag-transitive incidence geometry. This note proves the graph assertions only; it does not assert a source-wide result about incidence geometries or replace the separate proof identifying the full type-permuting group of that construction. Its distinction between edge transitivity under A^+ and under A is essential.

The graph was found as an eight-fold binary homology cover of a cubic graph on 112 vertices. That search is provenance, not a premise: the retained adjacency and permutations are sufficient to replay the entire proof. In particular, no completeness claim about the search is made. General permutation-group arithmetic, breadth-first traversal and neighbour-colour refinement are standard procedures; no historical novelty claim is made for these methods.

Seven controls reject a loop, a nonbijective proposed generator, a permutation not preserving adjacency, substitution of a proper subgroup, an incomplete star, the genuine swapping involutions of a six-cycle, and the weakened use of root-only colour refinement. The six-cycle control passes all preceding full-group and transitivity tests, so it specifically tests the final nonsplitting obstruction.

To replay the required certificate from an exported copy, run

```
python3 -B -O check.py --out fresh-result.json
```

There is no Lean formalization in this packet. The finite certificate is an ordinary executable proof by exhaustive verification; both its complete program and its mandatory data must be counted wherever this construction is used. Optional GAP/nauty/Bliss checks are separately inventoried as validation and cannot stand in for the required fullness argument.